Lecture 9 Compendium: GIS for Nutrient and Water Management

Module 3: Technologies of GIS in Crop Protection

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Compendium for Advanced Study

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1 Introduction: The Site-Specific Paradigm

Traditional agriculture treats a 50-hectare field as a homogenous unit ("Blanket Application"). **Site-Specific Crop Management (SSCM)** acknowledges that soil nutrient content and water holding capacity vary significantly within meters.

In this module, we explore how GIS serves as the analytical engine to quantify this variability and generate actionable **Prescription Maps**.

1.1 The Precision Cycle

1. **Measure:** Data acquisition (Soil sampling, Satellite imagery).
2. **Analyze:** Geostatistical interpolation to create continuous surfaces.
3. **Prescribe:** Decision making (Defining Management Zones).
4. **Actuate:** Variable Rate Application (VRA).

2 Geostatistics: From Points to Surfaces

Soil sampling is expensive. We cannot sample every square meter. We sample discrete points and **interpolate** the spaces in between.

2.1 Spatial Autocorrelation

The statistical foundation of interpolation is that "close samples are more likely to have similar values than distant samples" (Tobler's Law).

2.2 Interpolation Algorithms

2.2.1 1. Inverse Distance Weighting (IDW)

A deterministic method. The value at an unsampled location $\hat{Z}(x_0)$ is a weighted average of neighbors, where weight drops with distance d.

$$\hat{Z}(x_0) = \frac{\sum_{i=1}^{N} \lambda_i Z(x_i)}{\sum_{i=1}^{N} \lambda_i}, \quad \text{where } \lambda_i = \frac{1}{d_i^p} \tag{1}$$

* **Pros:** Computationally fast. * **Cons:** Cannot estimate prediction error; "Bullseye" effect around data points.

2.2.2 2. Kriging (The PhD Standard)

A stochastic method (BLUP - Best Linear Unbiased Predictor). Unlike IDW, Kriging considers the **spatial structure** of the data using a **Semivariogram**.

The Semivariance $\gamma(h)$: Measures the average dissimilarity between data points separated by a distance h (lag).

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2 \tag{2}$$

Key Semivariogram parameters:

- **Nugget:** Micro-scale variability (measurement error or variation at distances smaller than sampling interval).

- **Sill:** The total variance where the graph flattens out.
- **Range:** The distance at which spatial autocorrelation ceases.

2.3 Python: Semivariogram Modeling

Using Python to visualize spatial structure before interpolation.

```python
import numpy as np
import matplotlib.pyplot as plt

def theoretical_semivariogram(h, nugget, sill, range_val):
    """
    Spherical Model of Semivariance.
    h: distance lag
    """
    # If distance is greater than range, covariance is just the sill
    if h > range_val:
        return sill

    # Spherical model equation
    val = nugget + (sill - nugget) * ((1.5 * (h / range_val)) - (0.5 * (h / range_val)**3))
    return val

# Simulation Parameters
nugget = 0.1  # Measurement error
sill = 1.0    # Total variance
range_v = 50  # Meters (spatial correlation distance)

distances = np.linspace(0, 100, 100)
gammas = [theoretical_semivariogram(d, nugget, sill, range_v) for d in distances]

plt.figure(figsize=(10, 6))
plt.plot(distances, gammas, label='Spherical Model', color='blue')
plt.axvline(x=range_v, color='red', linestyle='--', label='Range (50m)')
plt.axhline(y=sill, color='green', linestyle='--', label='Sill (1.0)')
plt.title("Semivariogram: Quantifying Spatial Structure")
plt.xlabel("Distance Lag h (m)")
plt.ylabel("Semivariance \gamma(h)")
plt.legend()
plt.grid(True)
plt.show()
```

Listing 1: Modeling Spatial Dependence with Python

3 Remote Sensing for Water Management

While soil moisture sensors give point data, satellites give spatial coverage.

3.1 Physics of Water Absorption

Water is transparent in the Visible spectrum but opaque (absorptive) in the **Short-Wave Infrared (SWIR)**. * **Dry Vegetation:** High NIR reflectance, Moderate SWIR reflectance. * **Wet Vegetation:** High NIR reflectance, **Low SWIR reflectance** (Absorbed by water).

3.2 Normalized Difference Water Index (NDWI)

Proposed by Gao (1996).

$$NDWI = \frac{\rho_{NIR} - \rho_{SWIR}}{\rho_{NIR} + \rho_{SWIR}} \tag{3}$$

- **Values > 0.5:** No water stress (High water content).
- **Values < 0:** High water stress or bare soil.

Research Application: NDWI detects canopy water stress *before* it becomes visible to the human eye (or RGB drone). This allows for proactive irrigation.

4 Digital Soil Mapping (DSM)

Traditional soil mapping involves drawing polygons around sample points. DSM uses statistical inference to predict soil properties based on environmental covariates.

4.1 The SCORPAN Model

McBratney et al. (2003) formalized DSM as:

$$S = f(s, c, o, r, p, a, n) \tag{4}$$

Where Soil properties (S) are a function of:

- **s:** Soil (other properties, e.g., using EC to predict Salinity).
- **c:** Climate (Rainfall, Temperature).
- **o:** Organisms (Vegetation cover, NDVI).
- **r:** Relief (Topography, Slope, Aspect from DEM).
- **p:** Parent Material (Geology).
- **a:** Age (Time).
- **n:** Spatial position (Coordinates).

PhD Methodology: Instead of just Kriging soil samples (which uses only 'n'), we train a **Random Forest Regressor** using 'r' (Elevation), 'o' (NDVI), and 'c' (Rainfall) as predictors to map Soil Organic Carbon.

5 Variable Rate Technology (VRT)

VRT is the actuation layer. It adjusts the rate of input (fertilizer/water/pesticide) based on the GIS prescription map.

5.1 Workflow

1. **Management Zones:** GIS classifies the field into zones (e.g., High Potential vs. Low Potential). 2. **Prescription Map:** Assigns a target rate (kg/ha) to each zone. 3. **Controller:** The onboard computer reads the GPS location, looks up the target rate on the map, and adjusts the flow valve.

5.2 Standard: ISO 11783 (ISOBUS)

This standard ensures that a Case IH tractor can pull a John Deere sprayer and use a Trimble display. The prescription map is typically an XML-based Shapefile.

6 Case Study: Nitrogen Use Efficiency in Tanzania

Problem: Smallholder maize farmers in Iringa overuse Urea in sandy zones (leaching) and underuse it in clay zones (yield gap). **GIS Intervention:** 1. **Data:** Sentinel-2 Satellite data (NDRE band for Nitrogen). 2. **Analysis:** Calculated Nitrogen Nutrition Index (NNI). 3. **Zoning:** Created 3 zones (High, Medium, Low N requirement). 4. **Result:** Farmers applying VRT (even manually) saved 25% on fertilizer costs while maintaining yield.

7 Review Questions & Assignment

1. **Geostatistics:** You sample soil pH at 100m intervals. The variogram shows a "Range" of 60m. What does this imply about your sampling strategy? Is interpolation valid between the points?

2. **Physics:** Why is NDWI preferred over NDVI for irrigation scheduling? Under what conditions would NDVI give a "False Positive" for healthy crops when they are actually thirsty?

3. **Modeling:** Using the SCORPAN model, identify which remote sensing datasets (covariates) you would download to map Soil Erosion Risk in the Usambara Mountains.

References

1. McBratney, A. B., et al. (2003). *On digital soil mapping.* Geoderma.

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3. Oliver, M. A., & Webster, R. (2014). *Basic Steps in Geostatistics: The Variogram and Kriging.* Springer.